

IoT Wireless Weather Monitoring System

GROUP H

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INTRODUCTION

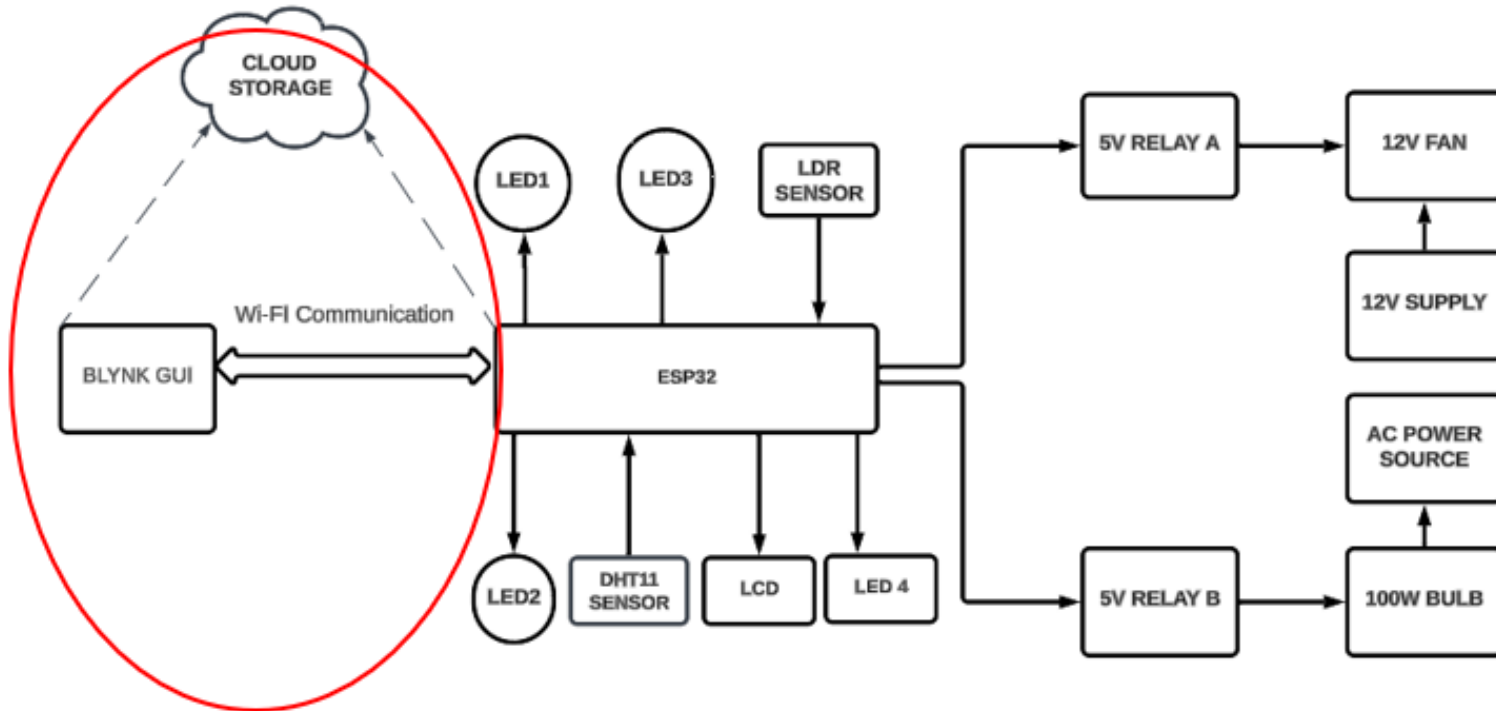
OBJECTIVES AND AIMS

- Developed Wi-Fi communication using ESP32 and Blynk
- Enabled cloud-based remote monitoring and control
- Integrated Automatic and Manual Modes for smart operation
- Used virtual pins for seamless interaction between hardware and app

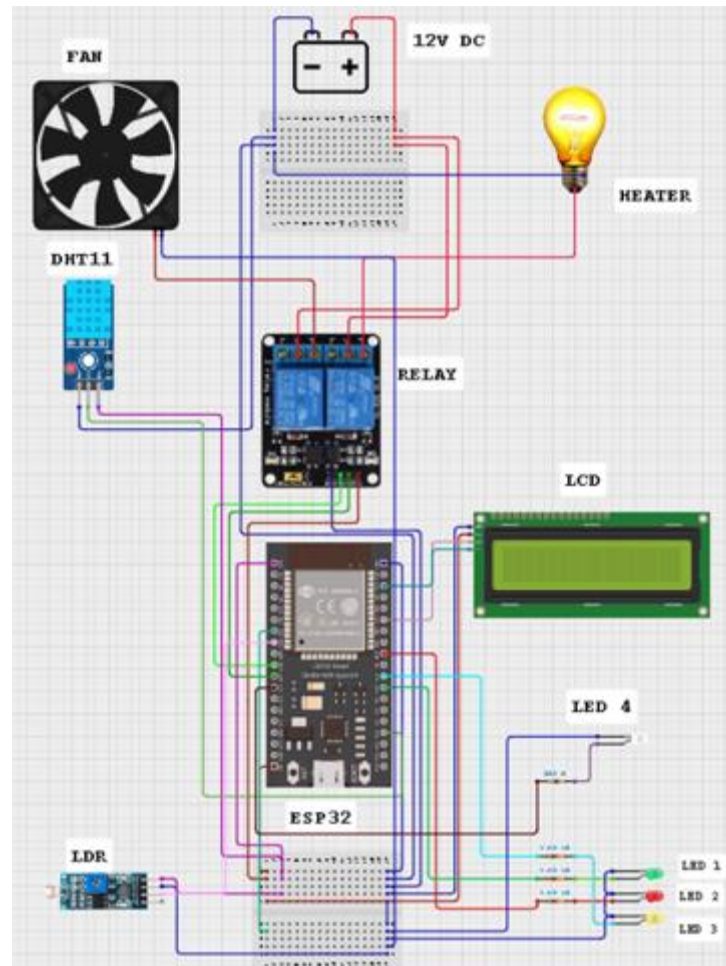
ESPECIFICATINS

- 5 V 2-Channel relay module was controlling fan and heating bulb.
- 3.3 V was powering ESP32 microcontroller, and LDR.
- 12 V power supply was powering 12VDC fan and AC power supply to power 100W light bulb

BLOCK DIAGRAM

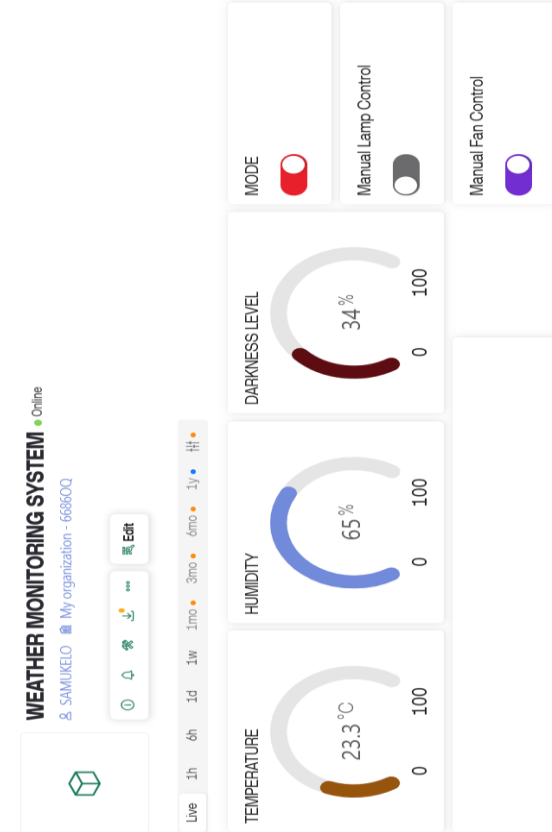
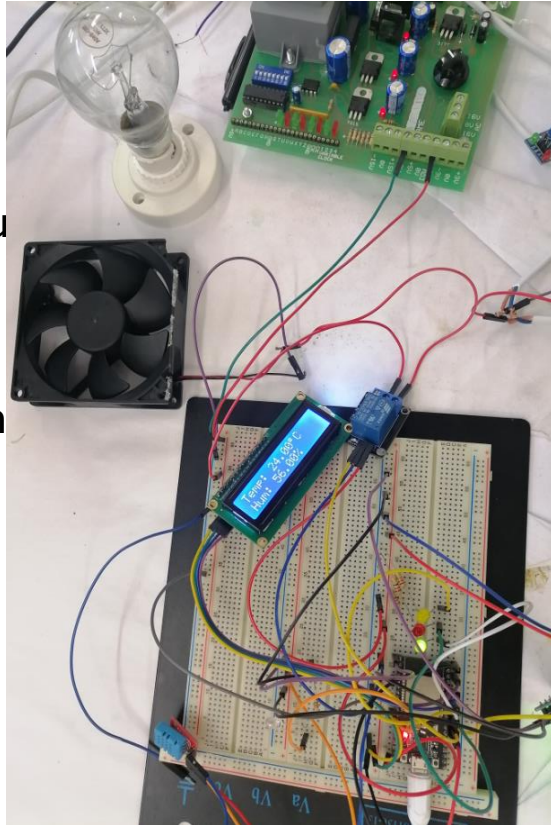


CIRCUIT DIAGRAM



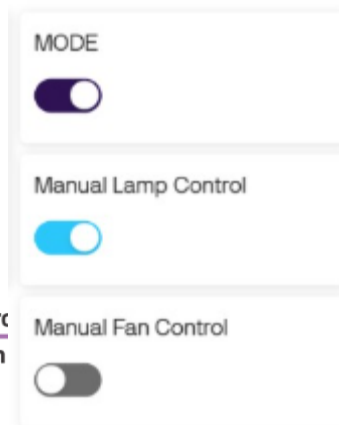
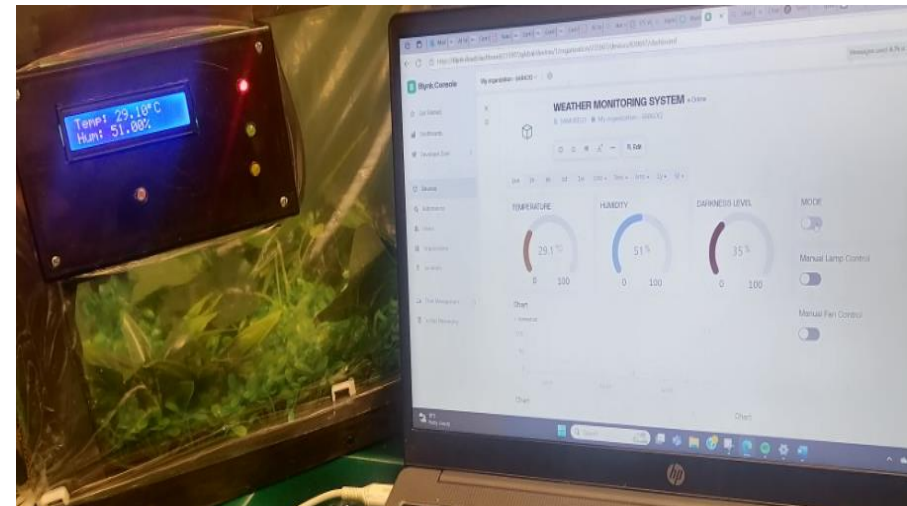
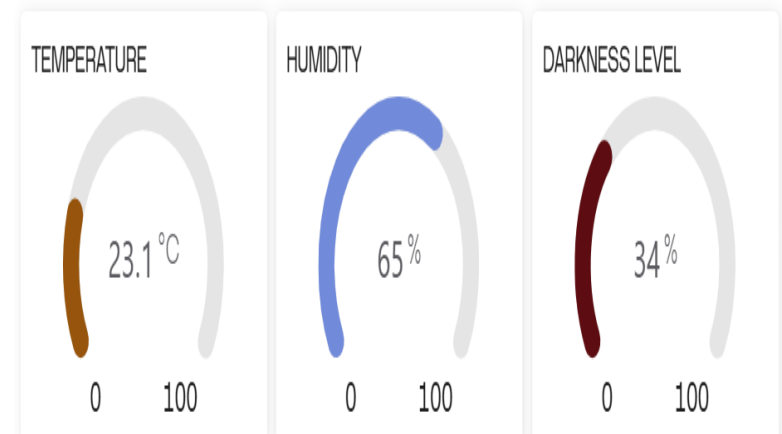
CONSTRUCTION

- Blynk switches set up for fan and bulb (V4,V5)
- Mode switch set to V3 (Auto/Manual toggle)
- Gauges created for V0,V1,V2 (temp, humidity, light)
- Final layout includes wiring and closed housing with ESP32, relays, and sensors



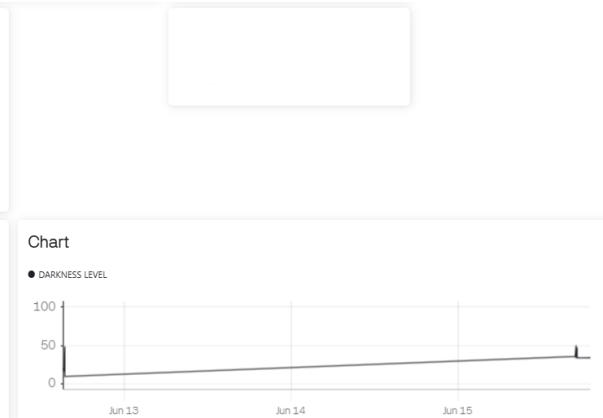
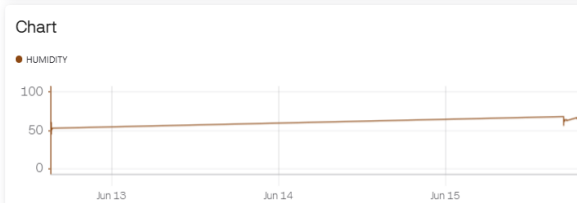
TESTING

- Verified Blynk GUI functionality
- Checked Manual control using virtual pins (V4,V5)
- Auto Mode tested with programmed temperature thresholds
- Blynk cloud logging tested for History Data accuracy

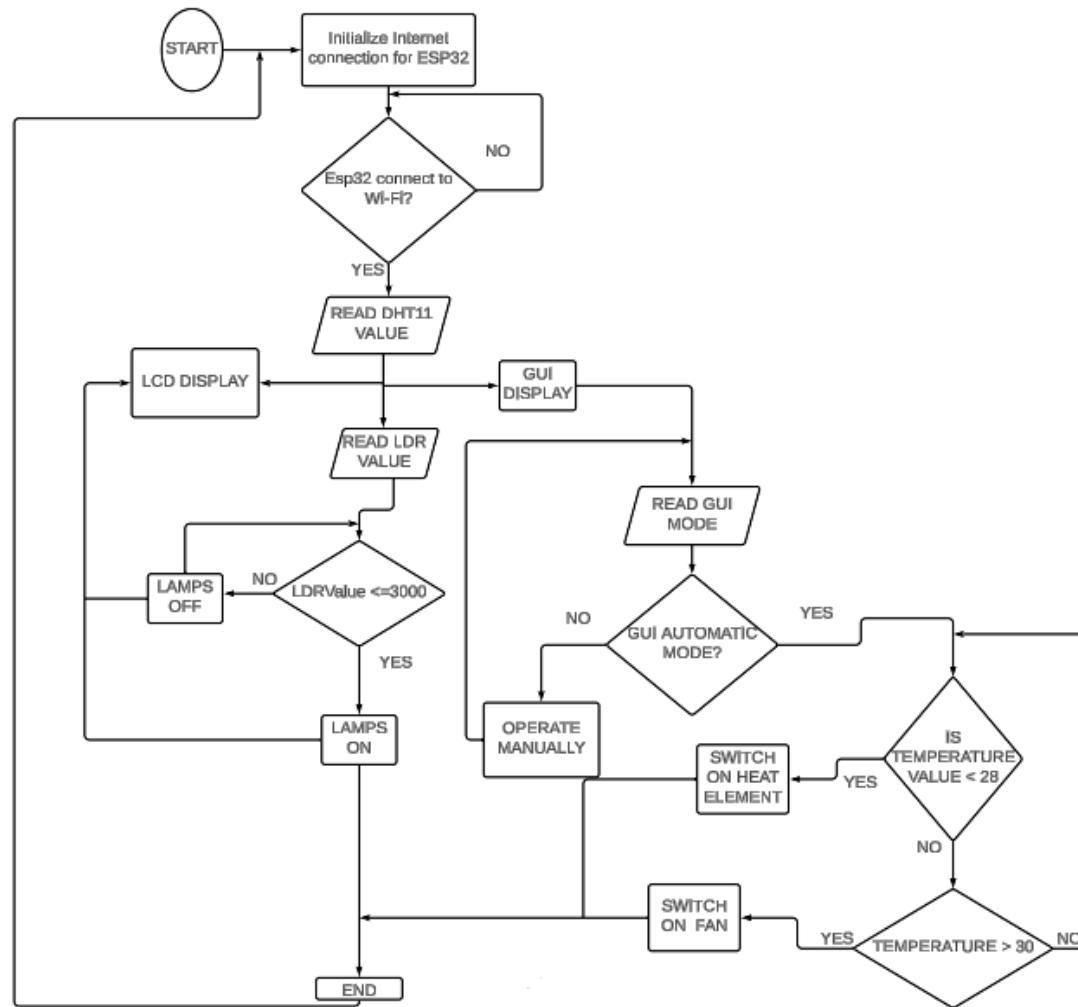


RESULTS

- Gauges displayed real-time sensor data
- Switches operated actuators as expected
- Auto Mode responded correctly to temperature limits
- Cloud storage captured historical data



FLOW CHART



LESSONS LEANT

- Improved skills in IoT and wireless communication
- Gained experience in virtual pin configuration and cloud setup
- Understood logic behind dual-mode systems
- Learned how to build responsive GUIs in Blynk

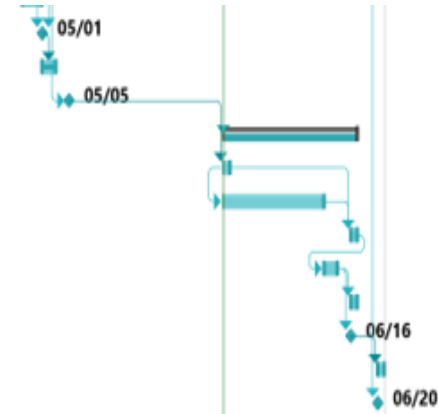
COMPLIANCES

- Complied with to instruction of the supervisor of not using a 12V lamp for a heating element, instead use a 100W light bulb.

PROJECT MANAGEMENT

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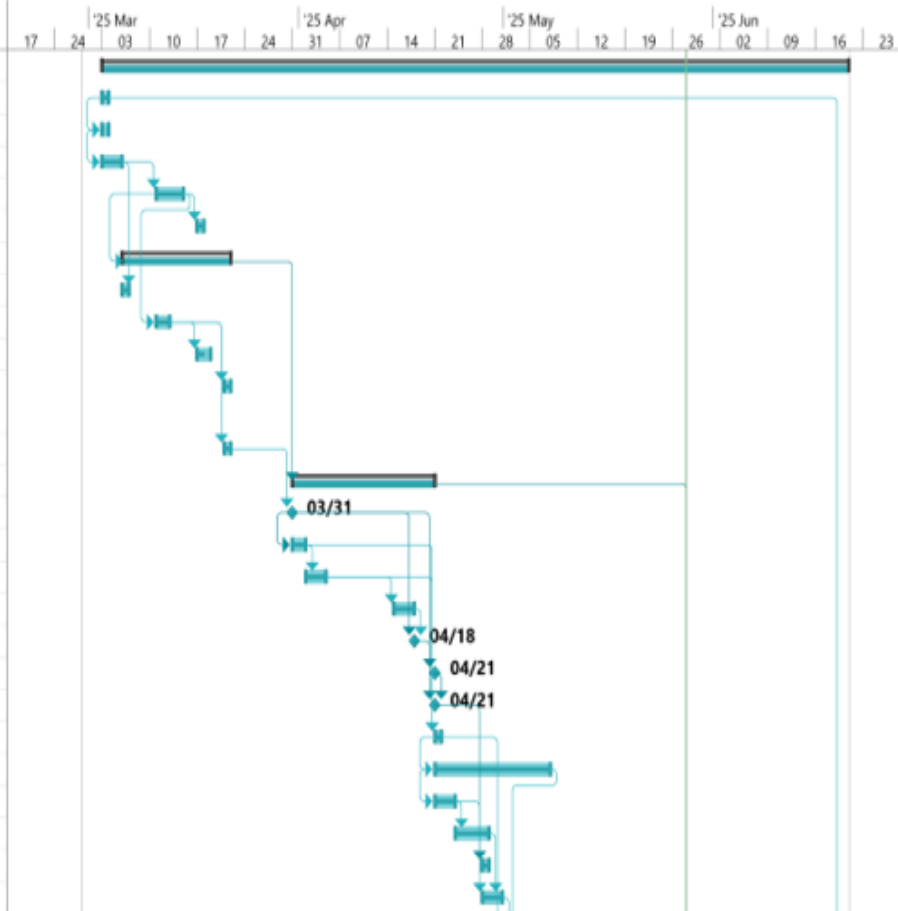
27	✓	✈	Finish Wi-Fi communication	0 days	Thu 25/05/01	Thu 25/05/01	21,26
28	✓	✈	Cloud Storage implementation	2 days	Thu 25/05/01	Fri 25/05/02	27
29	✓	✈	Finish wirelss communication	0 days	Mon 25/05/05	Mon 25/05/05	22
30		✈	Final report	14 days	Wed 25/05/28	Mon 25/06/16	13
31		✈	Start final individual report	1 day	Wed 25/05/28	Wed 25/05/28	29
32		✈	Research individual report	11 days	Wed 25/05/28	Wed 25/06/11	31SS
33		✈	Final report submission	1 day	Mon 25/06/16	Mon 25/06/16	31,32
34		✈	Presentation slide	2 days	Thu 25/06/12	Fri 25/06/13	33
35		✈	Presentation slides submission	1 day	Mon 25/06/16	Mon 25/06/16	34
36		✈	Presentation preparation	0 days	Mon 25/06/16	Mon 25/06/16	34
37		✈	Project presentation	1 day	Fri 25/06/20	Fri 25/06/20	36
38		✈	Finish project	0 days	Fri 25/06/20	Fri 25/06/20	2



Project: CHAPTER 4... Date: Wed 25/05/28	Task		Project Summary		Manual Task		Start-only		Deadline	
	Split		Inactive Task		Duration-only		Finish-only		Progress	
	Milestone		Inactive Milestone		Manual Summary Rollup		External Tasks		Manual Progress	
	Summary		Inactive Summary		Manual Summary		External Milestone			
Page 1										

PROJECT MANAGEMENT

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1	★	COMMUNICATION SYSTEM	80 days	Mon 25/03/03	Fri 25/06/20	
2	✓★	Start Project	1 day	Mon 25/03/03	Mon 25/03/03	
3	★	Group meetings	1 day	Mon 25/03/03	Mon 25/03/03	2SS
4	✓★	Topic allocation(IoT Engineer)	3 days	Mon 25/03/03	Wed 25/03/05	3SS
5	✓★	Communication research	4 days	Tue 25/03/11	Fri 25/03/14	4
6	✓★	Component Search	1 day	Mon 25/03/17	Mon 25/03/17	5
7	★	Final proposal	12 days	Thu 25/03/06	Fri 25/03/21	5SS
8	✓★	Task planning	1 day	Thu 25/03/06	Thu 25/03/06	4
9	✓★	Research communication method	2 days	Tue 25/03/11	Wed 25/03/12	5
10	★	Drawing circuit diagram	2 days	Mon 25/03/17	Tue 25/03/18	9
11	✓★	Combining communication proposal information	1 day	Fri 25/03/21	Fri 25/03/21	9
12	✓★	Group submission of the task	1 day	Fri 25/03/21	Fri 25/03/21	9
13	✓★	Project Progress Report	15 days	Mon 25/03/31	Mon 25/04/21	7
14	✓★	Start Progress report	0 days	Mon 25/03/31	Mon 25/03/31	12
15	✓★	Step-up BLYNK account	2 days	Mon 25/03/31	Tue 25/04/01	14SS
16	✓★	Set-up Blynk widget	3 days	Wed 25/04/02	Fri 25/04/04	15
17	✓★	Start progress presentation slide	3 days	Tue 25/04/15	Thu 25/04/17	16
18	✓★	Progress presentation Submission	0 days	Fri 25/04/18	Fri 25/04/18	14,17
19	✓★	Presentation	0 days	Mon 25/04/21	Mon 25/04/21	18
20	✓★	Finish progress	0 days	Mon 25/04/21	Mon 25/04/21	14,19
21	✓★	Start wireless communication	1 day	Mon 25/04/21	Mon 25/04/21	15,16
22	✓★	Wireless communication	13 days	Mon 25/04/21	Wed 25/05/07	21SS
23	✓★	Code ESP32 for Wi-Fi connection	3 days	Mon 25/04/21	Wed 25/04/23	22SS
24	✓★	Code ESP32 to send Data to BLYNK	3 days	Thu 25/04/24	Mon 25/04/28	23
25	✓★	Individual Progress Report Submission	1 day	Mon 25/04/28	Mon 25/04/28	20
26	✓★	Take results	3 days	Mon 25/04/28	Wed 25/04/30	23,24



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THANK YOU

QUESTIONS ?